Optimization for Data Science

Convex Optimization Problems

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Why convex optimization?

Optimization is hard in general:

- countless local minima
- saddle points and flat plateaus
- nonlinear constraints can create highly irregular feasible regions

Key message

When the objective and constraints are **convex**, the landscape simplifies dramatically:

- every local minimum is automatically a global minimum;
- near-optimal sets are convex;
- powerful algorithms solve many such problems efficiently.

Companion notebooks and chapter map

Companion notebooks

- Convex Optimization Examples
- Portfolio Optimization
- Optimal Control
- Logistic Regression

What we will cover

1. Standard form and convexity
2. Structural properties
3. First-order optimality
4. LP / QP / QCQP / SOCP / SDP
5. Practical modeling with CVXPY

General optimization problem: standard form

Standard form

$$
\begin{aligned}
\min_{x \in \mathbb{R}^n} \quad & f_0(x) \\
\text{s.t.} \quad & f_i(x) \leq 0, \qquad i = 1, \dots, m, \\
& h_i(x) = 0, \qquad i = 1, \dots, p.
\end{aligned}
$$

- $x \in \mathbb{R}^n$: **optimization / decision variable**
- f_0: **objective** (cost) function
- f_i: **inequality constraint** functions
- h_i: **equality constraint** functions

Unconstrained optimization

If $m = p = 0$, the problem reduces to

$$\min_{x \in \mathbb{R}^n} f_0(x).$$

Domain, feasible set, and optimal value

Definitions

$$\mathcal{D} = \bigcap_{i=0}^{m} \mathrm{dom}\, f_i \;\cap\; \bigcap_{i=1}^{p} \mathrm{dom}\, h_i.$$

A point $x_0 \in \mathcal{D}$ is **feasible** if

$$f_i(x_0) \le 0 \ (i \in [m]), \qquad h_i(x_0) = 0 \ (i \in [p]).$$

The **feasible set** is

$$\Omega = \{x \in \mathcal{D} : f_i(x) \le 0 \ \forall i,\ h_i(x) = 0 \ \forall i\}.$$

The **optimal value** is

$$f^* = \inf\{f_0(x) : x \in \Omega\}.$$

Two degenerate cases

- If $\Omega = \varnothing$ (infeasible), define $f^* = +\infty$.
- If feasible points can drive $f_0(x) \to -\infty$, define $f^* = -\infty$ (unbounded below).

Optimal points, local optimality, active constraints

- If x_0 is feasible and $f_0(x_0) = f^* > -\infty$, then x_0 is an **optimal point**.
- If an optimal point exists, the optimal value is **attained**; the problem is **solvable**.
- If x_0 is feasible and $f_0(x_0) \leq f^* + \varepsilon$, then x_0 is ε-**optimal**.
- If $f_i(x) = 0$ at a feasible point, then inequality constraint i is **active / binding**.

Local optimality

A feasible point x_0 is **locally optimal** if there exists $R > 0$ such that x_0 solves

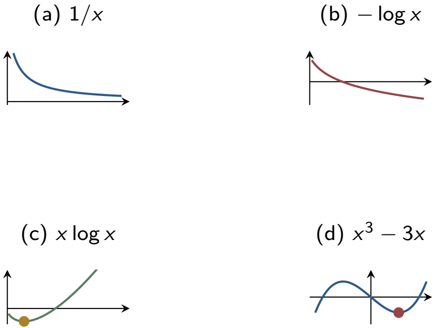
$$\min f_0(z) \quad \text{s.t.} \quad f_i(z) \leq 0,\ h_i(z) = 0,\ \|z - x_0\| \leq R.$$

Why this matters

In nonconvex optimization, local optimality is often the best algorithms can certify. In convex optimization, local optimality will turn out to be enough.

One-dimensional unconstrained optimization: four behaviors

- **Infimum not attained:** $f(x) = 1/x$ on $\mathbb{R}_{++}$ has $f^* = 0$, but no minimizer.
- **Unbounded below:** $f(x) = -\log x$ on $\mathbb{R}_{++}$ has $f^* = -\infty$.
- **Well-behaved minimum:** $f(x) = x \log x$ on $\mathbb{R}_{++}$ has unique minimizer $x^* = e^{-1}$ with $f^* = -e^{-1}$.
- **Local but not global:** $f(x) = x^3 - 3x$ has local minimum at $x = 1$, but $f(x) \to -\infty$ as $x \to -\infty$.



Worked calculation: minimizing $x \log x$ on $\mathbb{R}_{++}$

Let

$$f(x) = x \log x, \qquad x > 0.$$

Differentiate:

$$f'(x) = \log x + 1.$$

Set the derivative equal to zero:

$$\log x + 1 = 0 \qquad \Longrightarrow \qquad x = e^{-1}.$$

Differentiate again:

$$f''(x) = \frac{1}{x} > 0 \qquad \text{for all } x > 0.$$

So f is strictly convex on $\mathbb{R}_{++}$, which means any critical point is automatically the unique global minimizer.
Finally,

$$f(e^{-1}) = e^{-1} \log(e^{-1}) = -e^{-1}.$$

Takeaway

This is the clean case: the minimum exists, is attained, and is unique.

Worked calculation: a local minimum that is not global

Consider

$$f(x) = x^3 - 3x.$$

Differentiate:

$$f'(x) = 3x^2 - 3 = 3(x-1)(x+1).$$

So the critical points are

$$x = -1, \qquad x = 1.$$

Second derivative:

$$f''(x) = 6x.$$

Therefore

$$f''(-1) = -6 < 0 \Rightarrow x = -1 \text{ is a local maximum,}$$

$$f''(1) = 6 > 0 \Rightarrow x = 1 \text{ is a local minimum.}$$

But as $x \to -\infty$, $x^3 - 3x \to -\infty$, so the problem is unbounded below. Hence the local minimum at $x = 1$ is not global.

Feasibility problems

Feasibility problem

Does there exist x such that

$$f_i(x) \leq 0 \ \forall i, \qquad h_i(x) = 0 \ \forall i$$

???

Equivalent optimization view

This is the same as

$$\min 0 \quad \text{s.t.} \quad f_i(x) \leq 0, \ h_i(x) = 0.$$

- If the constraints are feasible, the trivial objective is attained and $f^* = 0$.
- If no feasible point exists, then $f^* = +\infty$.

Feasibility is just optimization with a constant objective.

Convex optimization in standard form

Convex optimization problem

A problem in standard form is **convex** if

1. the objective f_0 is convex,
2. each inequality function f_i is convex,
3. each equality constraint is **affine**: $h_i(x) = a_i^\top x - b_i$.

$$\begin{aligned} \min \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x) \le 0, \quad i = 1, \dots, m, \\ & a_i^\top x = b_i, \quad i = 1, \dots, p. \end{aligned}$$

Convex domain

$$\mathcal{D} = \bigcap_{i=0}^{m} \operatorname{dom} f_i,$$

so the domain is convex.

Why the feasible set of a convex program is convex

Recall the feasible set:

$$\Omega = \mathcal{D} \cap \bigcap_{i=1}^{m} \{x : f_i(x) \leq 0\} \cap \bigcap_{i=1}^{p} \{x : a_i^\top x = b_i\}.$$

- Because each f_i is convex, its sublevel set $\{x : f_i(x) \leq 0\}$ is convex.
- Each equality set $\{x : a_i^\top x = b_i\}$ is a hyperplane, hence convex.
- The domain $\mathcal{D}$ is convex.
- An intersection of convex sets is convex.

Mini-proof for one inequality constraint

If x, y satisfy $f_i(x) \leq 0$ and $f_i(y) \leq 0$, then for any $\lambda \in [0, 1]$,

$$f_i(\lambda x + (1 - \lambda)y) \leq \lambda f_i(x) + (1 - \lambda) f_i(y) \leq 0.$$

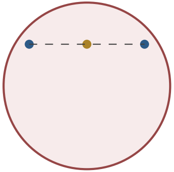
So convex combinations preserve feasibility.

Why only affine equality constraints?

Equality constraints are delicate

A nonlinear equality $h(x) = 0$ usually defines a curved level set, which is **not** convex.

- Example: $x_1^2 + x_2^2 = 1$ is a circle.
- The circle is not convex: the midpoint of two points on the circle usually lies inside it, not on it.
- If nonlinear equalities are allowed, the feasible set may fail to be convex.



$x_1^2 + x_2^2 = 1$ is not convex

Affine equalities are safe

Hyperplanes $a^\top x = b$ and intersections of hyperplanes are convex (indeed affine) sets.

Abstract view of convex optimization

$$\min_{x \in \Omega} f_0(x), \qquad \text{where } f_0 \text{ is convex and } \Omega \subseteq \mathbb{R}^n \text{ is convex.}$$

- The standard form is simply the most common way to describe the convex feasible set Ω.
- Convex inequality constraints contribute convex **sublevel sets**.
- Affine equalities contribute **hyperplanes**.
- Intersections of convex sets remain convex.

Feasible set decomposition

$$\Omega = \mathcal{D} \cap \bigcap_{i=1}^{m} \{x : f_i(x) \le 0\} \cap \bigcap_{i=1}^{p} \{x : a_i^\top x = b_i\}.$$

Three structural properties that change everything

For convex optimization, the geometry of the solution set becomes much simpler.

1. **ε-suboptimal sets are convex.**
2. **Strict convexity implies uniqueness.**
3. **Local minima are global minima.**

Big takeaway

These are exactly the properties that make convex problems reliable, interpretable, and algorithmically tractable.

Convex ε-suboptimal sets

Theorem

For any $\varepsilon \geq 0$, the set

$$S_\varepsilon = \{x \in \Omega : f_0(x) \leq f^* + \varepsilon\}$$

is convex.

Proof.

Take $x, y \in S_\varepsilon$ and $\lambda \in [0, 1]$. Since Ω is convex,

$$z = \lambda x + (1 - \lambda) y \in \Omega.$$

By convexity of f_0,

$$f_0(z) \leq \lambda f_0(x) + (1 - \lambda) f_0(y) \leq \lambda(f^* + \varepsilon) + (1 - \lambda)(f^* + \varepsilon) = f^* + \varepsilon.$$

Hence $z \in S_\varepsilon$. $\square$

Interpretation

If two feasible points are both near-optimal, then every convex combination of them is near-optimal too.

Proof details: convex ε-suboptimal sets

We want to show that if two points are ε-suboptimal, then everything between them is also ε-suboptimal.

1. Take any $x, y \in S_\varepsilon$. By definition,

$$x, y \in \Omega, \qquad f_0(x) \le f^* + \varepsilon, \qquad f_0(y) \le f^* + \varepsilon.$$

2. For any $\lambda \in [0, 1]$, define

$$z = \lambda x + (1 - \lambda) y.$$

   Since Ω is convex, $z \in \Omega$.

3. Apply convexity of f_0:

$$f_0(z) \le \lambda f_0(x) + (1 - \lambda) f_0(y).$$

4. Use the bounds on $f_0(x)$ and $f_0(y)$:

$$f_0(z) \le \lambda(f^* + \varepsilon) + (1 - \lambda)(f^* + \varepsilon) = f^* + \varepsilon.$$

So z is feasible and still within ε of optimal. Therefore S_ε is convex.

Strict convexity implies uniqueness

Theorem

If f_0 is **strictly convex**, then the optimal solution (if it exists) is unique.

Proof.

Suppose $x^* \neq y^*$ are both optimal with $f_0(x^*) = f_0(y^*) = f^*$. Since Ω is convex,

$$z = \tfrac{1}{2}x^* + \tfrac{1}{2}y^* \in \Omega.$$

Strict convexity gives

$$f_0(z) < \tfrac{1}{2}f_0(x^*) + \tfrac{1}{2}f_0(y^*) = f^*,$$

contradicting optimality of f^*. □

Lasso example

$$\min_{\beta} \|y - X\beta\|_2^2 \quad \text{s.t.} \quad \|\beta\|_1 \leq s.$$

Unique if X has full column rank; not necessarily unique when $n < d$.

Proof details: strict convexity implies uniqueness

Assume, for contradiction, that there are two distinct optimal points:
$x^* \neq y^*, \qquad f_0(x^*) = f_0(y^*) = f^*.$

1. Since the feasible set Ω is convex, their midpoint
$$z = \tfrac{1}{2}x^* + \tfrac{1}{2}y^*$$
is also feasible.
2. Because f_0 is strictly convex and $x^* \neq y^*$,
$$f_0(z) < \tfrac{1}{2}f_0(x^*) + \tfrac{1}{2}f_0(y^*) = \tfrac{1}{2}f^* + \tfrac{1}{2}f^* = f^*.$$
3. But f^* is the smallest objective value achievable by any feasible point, so a feasible point cannot have value smaller than f^*.

This contradiction shows that two distinct minimizers cannot exist.

Contrast with ordinary convexity

If f_0 is convex but not strictly convex, the set of minimizers is still convex, but it may be a whole line segment or flat face.

Local minimum = global minimum

Theorem

Any local optimal point of a convex optimization problem is also globally optimal.

- Suppose x_0 is locally optimal but not globally optimal.
- Then there exists a feasible point y with $f_0(y) < f_0(x_0)$.
- Because the feasible set is convex, the entire segment

$$z_\lambda = \lambda x_0 + (1-\lambda)y, \qquad \lambda \in [0,1],$$

  stays feasible.
- By convexity of f_0,

$$f_0(z_\lambda) \leq \lambda f_0(x_0) + (1-\lambda) f_0(y) < f_0(x_0), \qquad \lambda \in (0,1).$$

Contradiction

For λ close enough to 1, the point z_λ lies arbitrarily close to x_0, violating local optimality.

Proof details I: a better feasible point creates a better segment

Assume x_0 is locally optimal but not globally optimal. Then two facts hold simultaneously:

- there exists $R > 0$ such that

$$z \in \Omega,\ \|z - x_0\| \le R \quad \Longrightarrow \quad f_0(z) \ge f_0(x_0),$$

  because x_0 is locally optimal;

- there exists some feasible point $y \in \Omega$ with

$$f_0(y) < f_0(x_0),$$

  because x_0 is not globally optimal.

Now form the segment

$$z_\lambda = \lambda x_0 + (1-\lambda)y, \qquad \lambda \in [0,1].$$

Since Ω is convex, every z_λ is feasible. By convexity of f_0,

$$f_0(z_\lambda) \le \lambda f_0(x_0) + (1-\lambda) f_0(y) < f_0(x_0)$$

for every $\lambda \in (0,1)$.

Proof details II: make the better point arbitrarily close to x_0

We now compute the distance from z_λ to x_0:

$$z_\lambda - x_0 = \lambda x_0 + (1-\lambda)y - x_0 = (1-\lambda)(y - x_0).$$

Taking norms gives

$$\|z_\lambda - x_0\| = (1-\lambda)\,\|y - x_0\|\,.$$

So by choosing λ sufficiently close to 1, we can force

$$\|z_\lambda - x_0\| \le R.$$

That means z_λ lies inside the neighborhood where local optimality says no feasible point can do better than x_0. But from the previous slide we already know

$$f_0(z_\lambda) < f_0(x_0).$$

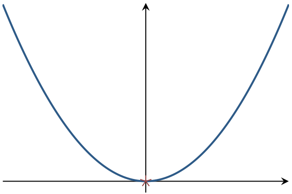
This is a contradiction.

Conclusion

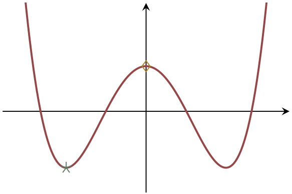
A convex problem cannot have a point that is only locally optimal. Every local minimum must already be global.

Convex vs nonconvex landscapes

Convex: $f(x) = x^2$



Nonconvex: $x^4 - 3x^2 + 1$



- Convex problems have no deceptive local minima.
- Nonconvex problems can have multiple local optima, saddle points, and disconnected regions of good solutions.

Optimality conditions for differentiable problems

When the objective is differentiable, convexity turns gradient conditions into **global certificates of optimality**.

1. **Unconstrained:** $\nabla f_0(x^*) = 0$.
2. **Affine equalities:** $\nabla f_0(x^*) \perp \mathcal{N}(A)$.
3. **General convex set Ω:**

$$\nabla f_0(x^*)^\top (y - x^*) \geq 0 \quad \forall y \in \Omega.$$

Unconstrained optimality

Theorem

If f_0 is convex and differentiable on $\mathbb{R}^n$, then

$$x^* \in \arg\min_{x \in \mathbb{R}^n} f_0(x) \qquad \Longleftrightarrow \qquad \nabla f_0(x^*) = 0.$$

- In ordinary calculus, $\nabla f_0(x^*) = 0$ is a necessary condition.
- For **convex** f_0, it becomes necessary *and* sufficient.

Reason

The first-order inequality for convex functions says

$$f_0(y) \geq f_0(x^*) + \nabla f_0(x^*)^\top (y - x^*).$$

If $\nabla f_0(x^*) = 0$, then $f_0(y) \geq f_0(x^*)$ for all y.

Proof details: unconstrained first-order optimality

Sufficiency: $\nabla f_0(x^*) = 0$ implies optimality

For a differentiable convex function,

$$f_0(y) \geq f_0(x^*) + \nabla f_0(x^*)^\top (y - x^*) \quad \forall y.$$

If $\nabla f_0(x^*) = 0$, this becomes

$$f_0(y) \geq f_0(x^*) \quad \forall y.$$

So no point has smaller objective value, and x^* is globally optimal.

Necessity: optimality implies $\nabla f_0(x^*) = 0$

If x^* is optimal, then for every direction d and small $t > 0$,

$$f_0(x^* + td) \geq f_0(x^*).$$

Subtract $f_0(x^*)$, divide by t, and let $t \downarrow 0$:

$$\nabla f_0(x^*)^\top d \geq 0 \quad \forall d.$$

Apply the same argument to $-d$ to get the reverse inequality. Hence $\nabla f_0(x^*)^\top d = 0$ for every direction d, so $\nabla f_0(x^*) = 0$.

Equality-constrained optimality

Problem

$$\min f_0(x) \qquad \text{s.t.} \quad Ax = b,$$

where f_0 is convex and differentiable.

Theorem

A feasible point x^* is optimal iff

$$\nabla f_0(x^*) \perp \mathcal{N}(A),$$

where $\mathcal{N}(A) = \{v : Av = 0\}$ is the null space of A.

- Feasible perturbations must satisfy $A(x^* + tv) = b$, so $v \in \mathcal{N}(A)$.
- At optimum, the gradient can have no descent component along feasible directions.

Geometric meaning

The gradient points orthogonally to the affine subspace of feasible directions.

Deriving the equality-constrained condition I: feasible directions

For the problem

$$\min f_0(x) \qquad \text{s.t.} \quad Ax = b,$$

start from a feasible point x^* and perturb it to $x^* + tv$. For this new point to remain feasible, we need

$$A(x^* + tv) = b.$$

Since $Ax^* = b$, subtracting gives

$$tAv = 0 \qquad \Longrightarrow \qquad Av = 0.$$

So the feasible directions are exactly the vectors in the null space

$$\mathcal{N}(A) = \{v : Av = 0\}.$$

Interpretation

Under equality constraints, you are not free to move in arbitrary directions. You can only move along directions tangent to the affine constraint set.

Deriving the equality-constrained condition II: orthogonality

At an optimum x^*, no feasible direction $v \in \mathcal{N}(A)$ can decrease the objective. If there were some $v \in \mathcal{N}(A)$ with

$$\nabla f_0(x^*)^\top v < 0,$$

then for small $t > 0$ the feasible point $x^* + tv$ would satisfy

$$f_0(x^* + tv) \approx f_0(x^*) + t\nabla f_0(x^*)^\top v < f_0(x^*),$$

contradicting optimality. Therefore

$$\nabla f_0(x^*)^\top v = 0 \qquad \forall v \in \mathcal{N}(A),$$

which is exactly

$$\nabla f_0(x^*) \perp \mathcal{N}(A).$$

Deriving the equality-constrained condition III: multiplier form

The orthogonality condition has a useful algebraic reformulation. Because the orthogonal complement of the null space is the range of $A^\top$,

$$\mathcal{N}(A)^\perp = \operatorname{range}(A^\top).$$

Therefore

$$\nabla f_0(x^*) \perp \mathcal{N}(A) \quad \iff \quad \nabla f_0(x^*) \in \operatorname{range}(A^\top).$$

Equivalently, there exists a vector λ such that

$$\nabla f_0(x^*) = A^\top \lambda.$$

Why this matters

This is the equality-constrained version of the familiar Lagrange-multiplier condition.

General constrained optimality

Theorem

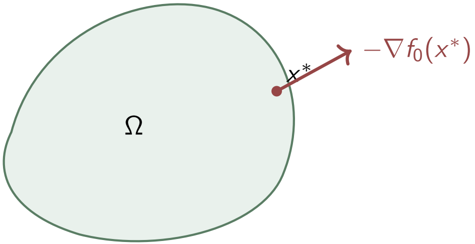
If f_0 is convex and differentiable, then x^* solves

$$\min_{x \in \Omega} f_0(x)$$

iff

$$\nabla f_0(x^*)^\top (y - x^*) \geq 0 \qquad \forall y \in \Omega.$$

- **Sufficiency:** combine convexity with the inequality above.
- **Necessity:** otherwise a feasible direction would decrease the objective.
- This generalizes both the unconstrained and equality-constrained cases.



Proof details I: why the constrained inequality is sufficient

Assume

$$\nabla f_0(x^*)^\top (y - x^*) \geq 0 \qquad \forall y \in \Omega.$$

For a differentiable convex function, the first-order inequality says

$$f_0(y) \geq f_0(x^*) + \nabla f_0(x^*)^\top (y - x^*) \qquad \forall y \in \Omega.$$

Combine the two inequalities:

$$f_0(y) \geq f_0(x^*) \qquad \forall y \in \Omega.$$

So no feasible point has smaller objective value than x^*, and x^* is optimal.

What happened conceptually?

The gradient gives a supporting hyperplane to a convex function. If that hyperplane already slopes upward in every feasible direction, then the function itself cannot go down along the feasible set.

Proof details II: why the constrained inequality is necessary

Suppose x^* is optimal, but there exists some feasible point $y \in \Omega$ with

$$\nabla f_0(x^*)^\top (y - x^*) < 0.$$

For small $t > 0$, define the point on the feasible segment

$$x_t = x^* + t(y - x^*) = (1 - t)x^* + ty.$$

Because Ω is convex, $x_t \in \Omega$. By differentiability,

$$f_0(x_t) = f_0(x^*) + t\nabla f_0(x^*)^\top (y - x^*) + o(t).$$

The coefficient of t is negative, so for all sufficiently small $t > 0$,

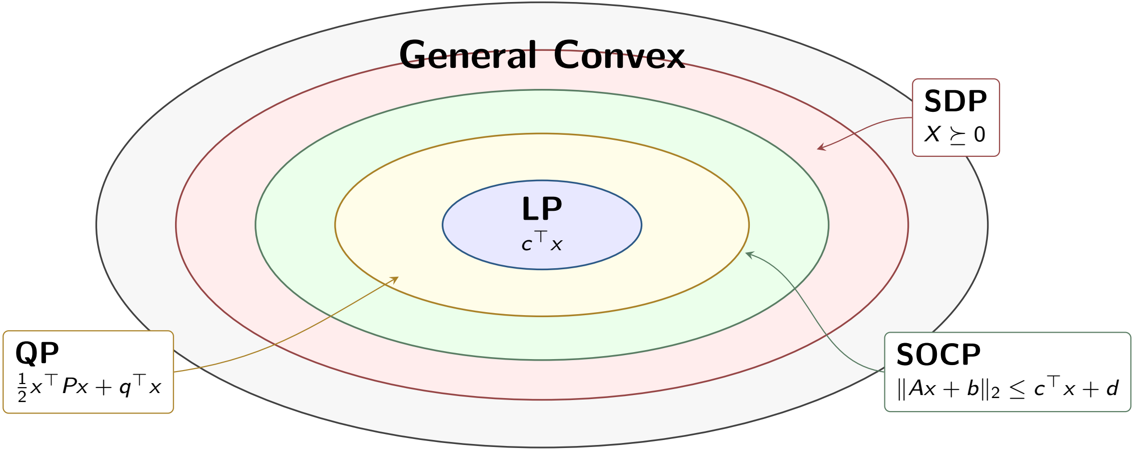
$$f_0(x_t) < f_0(x^*),$$

which contradicts optimality.

Conclusion

At an optimum, every feasible direction has nonnegative directional derivative.

Hierarchy of convex programs



$$\text{LP} \subseteq \text{QP} \subseteq \text{SOCP} \subseteq \text{SDP} \subseteq \text{General Convex}.$$

1. Linear programs (LP)

Definition

A **linear program** has a linear objective and linear constraints:

$$\min \; c^\top x + d \quad \text{s.t.} \quad Gx \le h, \; Ax = b.$$

Special forms:

- standard form: $\min c^\top x$ s.t. $Ax = b$
- canonical form: $\min c^\top x$ s.t. $Ax \le b, \; x \ge 0$

- Feasible sets are **polyhedra**.
- Solutions occur at vertices for generic LPs.
- LPs appear in transportation, scheduling, network flow, and resource allocation.

LP examples: transportation and max flow

Transportation problem

$$\min_{x_{ij}} \sum_{i,j} c_{ij} x_{ij}$$

subject to

$$\sum_j x_{ij} \le s_i, \qquad \sum_i x_{ij} \ge d_j, \qquad x_{ij} \ge 0.$$

- s_i: supply
- d_j: demand
- x_{ij}: shipped quantity

Maximum flow

$$\max \sum_{v:(1,v)\in E} x_{1v}$$

subject to

$$0 \le x_{uv} \le C_{uv},$$

$$\sum_{u:(u,v)\in E} x_{uv} = \sum_{w:(v,w)\in E} x_{vw}$$

for internal nodes v.

Both are LPs: linear objective + linear conservation/capacity constraints.

Linear classification and a brief history of LP

Separating hyperplane

Finding (w, b) such that

$$y_i(w^\top x_i + b) \geq 1, \qquad i = 1, \ldots, m,$$

is a **feasibility LP**.

The max-margin version,

$$\min_{w,b} \ \frac{1}{2}\|w\|_2^2 \quad \text{s.t.} \quad y_i(w^\top x_i + b) \geq 1,$$

is a **QP**.

Historical milestones

- Fourier-Motzkin elimination
- Kantorovich and Koopmans: resource allocation and transport
- Dantzig: simplex method (1947)
- Khachiyan: ellipsoid method (1979)
- Karmarkar: interior-point method (1984)
- Spielman-Teng: smoothed analysis of simplex (2001)

2. Quadratic programs (QP)

Definition

A **quadratic program** minimizes a convex quadratic objective under linear constraints:

$$\min_x \ \frac{1}{2}x^\top P x + q^\top x + r \quad \text{s.t.} \quad Gx \leq h, \ Ax = b,$$

with $P \succeq 0$.

- $P \succeq 0$ guarantees convexity.
- If $P \succ 0$, the objective is strictly convex, so the solution is unique (when feasible).
- LP is the special case $P = 0$.

Geometry

LP contours are linear; QP contours are ellipsoids. Therefore the optimum of a QP need not occur at a vertex of the feasible polyhedron.

Why a quadratic objective is convex when $P \succeq 0$

For

$$f(x) = \frac{1}{2}x^\top Px + q^\top x + r,$$

the gradient and Hessian are

$$\nabla f(x) = Px + q, \qquad \nabla^2 f(x) = P.$$

A twice-differentiable function is convex exactly when its Hessian is positive semidefinite everywhere, so here the condition is simply

$$P \succeq 0.$$

Equivalently,

$$v^\top Pv \geq 0 \qquad \forall v \in \mathbb{R}^n.$$

Interpretation

P controls the curvature of the quadratic bowl. If $P \succ 0$, the bowl curves upward in every direction, making the objective strictly convex.

QP examples

Constrained least squares

$$\min \|Ax - b\|_2^2 \quad \text{s.t.} \quad x \geq 0.$$

Equivalent QP with

$$P = 2A^\top A, \qquad q = -2A^\top b.$$

Markowitz portfolio

$$\min_x x^\top \Sigma x$$

$$\text{s.t. } \mu^\top x \geq r_{\min},\ \mathbf{1}^\top x = 1,\ x \geq 0.$$

Risk is quadratic, constraints are linear.

Distance between polyhedra

$$\min_{x_1, x_2} \ \|x_1 - x_2\|_2^2$$

$$\text{s.t. } A_1 x_1 \leq b_1,\ A_2 x_2 \leq b_2.$$

Closest points solve a QP.

Derivation: constrained least squares is a QP

Start from

$$\min_x \|Ax - b\|_2^2 \qquad \text{s.t.} \quad x \geq 0.$$

Expand the square:

$$\|Ax - b\|_2^2 = (Ax - b)^\top (Ax - b).$$

Multiply out:

$$(Ax - b)^\top (Ax - b) = x^\top A^\top Ax - 2b^\top Ax + b^\top b.$$

Now match the standard QP form:

$$\frac{1}{2} x^\top Px + q^\top x + r.$$

So we identify

$$P = 2A^\top A, \qquad q = -2A^\top b, \qquad r = b^\top b.$$

Why is $P \succeq 0$?

For any vector v,

$$v^\top Pv = 2v^\top A^\top Av = 2\,\|Av\|_2^2 \geq 0.$$

So the objective is convex, and the constraint $x \geq 0$ is linear.

3. Quadratically constrained QP (QCQP)

Definition

A **QCQP** minimizes a convex quadratic objective with convex quadratic inequality constraints:

$$\begin{aligned} \min_{x} \ & \tfrac{1}{2}x^\top P_0 x + q_0^\top x + r_0 \\ \text{s.t. } & \tfrac{1}{2}x^\top P_i x + q_i^\top x + r_i \leq 0, \quad i = 1, \dots, m, \\ & Ax = b, \end{aligned}$$

where $P_0, P_1, \dots, P_m \succeq 0$.

- QP is the special case with no quadratic inequality constraints.
- When $P_i \succ 0$, the set $\{x : \frac{1}{2}x^\top P_i x + q_i^\top x + r_i \leq 0\}$ is ellipsoidal.

QCQP examples

Trust-region subproblem

$$\min_{s}\ f(x) + \nabla f(x)^\top s + \tfrac{1}{2} s^\top \nabla^2 f(x) s$$

$$\text{s.t.}\quad s^\top s \le \Delta^2.$$

One quadratic objective + one quadratic constraint.

Portfolio with factor-risk bound

$$\min_{w}\ w^\top \Sigma w$$

$$\text{s.t. } \mu^\top w \ge r_{\min},\ \ \|Fw\|_2^2 \le \text{risk}_{\max},$$

$$0 \le w_i \le 0.1, \quad \mathbf{1}^\top w = 1.$$

Since $\|Fw\|_2^2 = w^\top F^\top F w$, this is a convex quadratic constraint.

4. Second-order cone programs (SOCP)

Definition

A **second-order cone program** has the form

$$\begin{aligned} \min_x \quad & f^\top x \\ \text{s.t.} \quad & \|A_i x + b_i\|_2 \le c_i^\top x + d_i, \quad i = 1, \ldots, m, \\ & Fx = g. \end{aligned}$$

- The constraint $\|u\|_2 \le t$ defines the second-order cone

$$\mathcal{C}_{n+1} = \{(u, t) : \|u\|_2 \le t\}.$$

- SOCP generalizes LP, QP, and QCQP.
- Norm constraints are the natural language of robustness and geometry.

SOCP examples

Robust LP

If uncertain coefficients satisfy

$$a_i \in \{\bar{a}_i + P_i u : \|u\|_2 \leq 1\},$$

then worst-case feasibility becomes

$$\bar{a}_i^\top x + \left\|P_i^\top x\right\|_2 \leq b_i.$$

Smallest enclosing circle

$$\min_{c_x, c_y, R} \quad R$$

$$\text{s.t. } \sqrt{(x_i - c_x)^2 + (y_i - c_y)^2} \leq R,$$

for each data point.

Robust classification

Worst-case perturbation balls lead to

$$\rho \|w\|_2 \leq y_i(w^\top x_i + b) - 1,$$

revealing max-margin SVM as robustness maximization.

Derivation I: robust LP as a worst-case linear inequality

Suppose the uncertain coefficient vector satisfies

$$a_i \in \{\bar{a}_i + P_i u : \|u\|_2 \leq 1\}.$$

The robust constraint requires

$$a_i^\top x \leq b_i \qquad \forall a_i \text{ in that set.}$$

Substitute the parameterization of a_i:

$$(\bar{a}_i + P_i u)^\top x \leq b_i \qquad \forall \|u\|_2 \leq 1.$$

This is equivalent to asking that the worst-case left-hand side still be below b_i:

$$\sup_{\|u\|_2 \leq 1} (\bar{a}_i + P_i u)^\top x \leq b_i.$$

Now separate nominal and uncertain parts:

$$\bar{a}_i^\top x + \sup_{\|u\|_2 \leq 1} u^\top P_i^\top x \leq b_i.$$

Derivation II: evaluating the worst case gives a norm

We now evaluate

$$\sup_{\|u\|_2 \le 1} u^\top P_i^\top x.$$

By Cauchy-Schwarz,

$$u^\top P_i^\top x \le \|u\|_2 \left\|P_i^\top x\right\|_2 \le \left\|P_i^\top x\right\|_2 .$$

Equality is attained when u points in the same direction as $P_i^\top x$. Therefore

$$\sup_{\|u\|_2 \le 1} u^\top P_i^\top x = \left\|P_i^\top x\right\|_2 .$$

Substituting back, the robust constraint becomes

$$\bar{a}_i^\top x + \left\|P_i^\top x\right\|_2 \le b_i,$$

which is exactly a second-order cone constraint.

Modeling lesson

A linear constraint with ellipsoidal uncertainty becomes a norm-bounded cone constraint.

Derivation: robust classification leads to the max-margin SVM

Require correct classification for every perturbation u with $\|u\|_2 \leq \rho$:

$$y_i(w^\top(x_i + u) + b) \geq 1.$$

Expand the uncertain term:

$$y_i(w^\top x_i + b) + y_i w^\top u \geq 1.$$

The worst case makes the linear term in u as small as possible. Since $y_i \in \{\pm 1\}$,

$$\inf_{\|u\|_2 \leq \rho} y_i w^\top u = -\rho \|w\|_2 .$$

Hence the robust constraint is equivalent to

$$y_i(w^\top x_i + b) - \rho \|w\|_2 \geq 1.$$

If we rescale so that $\rho \|w\|_2 = 1$, then maximizing robustness radius is equivalent to

$$\min \|w\|_2 \qquad \text{s.t.} \quad y_i(w^\top x_i + b) \geq 1,$$

which is the max-margin SVM formulation.

5. Semidefinite programs (SDP)

Definition

An **SDP** optimizes a linear objective over positive semidefinite matrices:

$$\min_{X \in \mathbb{S}^n} \ \langle C, X \rangle$$

$$\text{s.t.} \quad \langle A_i, X \rangle = b_i, \quad i = 1, \ldots, m, \qquad X \succeq 0.$$

- $X \succeq 0$ means all eigenvalues of X are nonnegative.
- LP constrains a vector to be elementwise nonnegative; SDP constrains a matrix to be PSD.
- SDP generalizes SOCP and therefore all previous classes in the hierarchy.

SDP duality and Slater's condition

Primal-dual pair

For the primal SDP

$$\min_{X} \ \langle C, X\rangle \quad \text{s.t.} \quad \langle A_i, X\rangle = b_i,\ X \succeq 0,$$

the dual is

$$\max_{y} \ b^\top y \quad \text{s.t.} \quad \sum_{i=1}^{m} y_i A_i \preceq C.$$

Important caveat

Unlike LP, strong duality does *not* hold automatically for SDP.

Slater's condition

If there exists a strictly feasible point $X \succ 0$ satisfying all equality constraints, then strong duality holds and there is no primal-dual gap.

SDP example: computing the largest eigenvalue

For a symmetric matrix Σ,

$$\lambda_{\max}(\Sigma) = \max_{\|x\|_2=1} x^\top \Sigma x.$$

This is nonconvex because $\|x\|_2 = 1$ is not a convex constraint.

Lift-and-relax idea

Set $X = xx^\top$. Then

$$\max_X \ \langle \Sigma, X \rangle \quad \text{s.t.} \quad \langle I, X \rangle = 1,\ X = xx^\top.$$

Dropping the rank-one constraint gives the SDP

$$\max_X \ \langle \Sigma, X \rangle \quad \text{s.t.} \quad \langle I, X \rangle = 1,\ X \succeq 0.$$

Equivalent SDP form

$$\lambda_{\max}(\Sigma) = \min_t \ t \quad \text{s.t.} \quad \Sigma \preceq tI.$$

Derivation: from the Rayleigh quotient to an SDP

For symmetric Σ,

$$\lambda_{\max}(\Sigma) = \max_{\|x\|_2=1} x^\top \Sigma x.$$

Introduce the lifted matrix

$$X = xx^\top.$$

Then

$$x^\top \Sigma x = \operatorname{tr}(x^\top \Sigma x) = \operatorname{tr}(\Sigma x x^\top) = \operatorname{tr}(\Sigma X) = \langle \Sigma, X \rangle .$$

The unit-norm constraint becomes

$$\operatorname{tr}(X) = \operatorname{tr}(xx^\top) = x^\top x = 1.$$

Also $X \succeq 0$ and $\operatorname{rank}(X) = 1$. So the problem is equivalent to

$$\max_X \ \langle \Sigma, X \rangle \qquad \text{s.t.} \quad \langle I, X \rangle = 1, \ X \succeq 0, \ \operatorname{rank}(X) = 1.$$

Dropping the rank-one constraint gives a convex SDP relaxation.

Why the eigenvalue SDP relaxation is tight

Let $v_{\max}$ be a unit eigenvector of Σ associated with $\lambda_{\max}(\Sigma)$. Then

$$X = v_{\max} v_{\max}^\top$$

is feasible for the relaxed SDP, since $X \succeq 0$ and $\mathrm{tr}(X) = 1$. Its objective value is

$$\langle \Sigma, X \rangle = v_{\max}^\top \Sigma v_{\max} = \lambda_{\max}(\Sigma).$$

Conversely, any feasible $X \succeq 0$ with $\mathrm{tr}(X) = 1$ admits an eigen-decomposition

$$X = \sum_k \alpha_k v_k v_k^\top, \qquad \alpha_k \geq 0, \qquad \sum_k \alpha_k = 1.$$

Therefore

$$\langle \Sigma, X \rangle = \sum_k \alpha_k v_k^\top \Sigma v_k \leq \lambda_{\max}(\Sigma) \sum_k \alpha_k = \lambda_{\max}(\Sigma).$$

So the SDP optimum is exactly $\lambda_{\max}(\Sigma)$.

Hierarchy summary table

Type	Objective	Constraints	Typical examples
LP	$c^\top x$	$Ax \le b,\ Ax = b$	transportation, max flow
QP	$\frac{1}{2}x^\top Px + q^\top x$	linear	portfolio, constrained LS
QCQP	convex quadratic	convex quadratic + affine	trust region, risk-bounded portfolio
SOCP	often linear	$\|A_i x + b_i\|_2 \le c_i^\top x + d_i$	robust LP, enclosing circle
SDP	$\langle C, X\rangle$	$\langle A_i, X\rangle = b_i,\ X \succeq 0$	eigenvalue, relaxations

$$\text{LP} \subset \text{QP} \subset \text{QCQP} \subset \text{SOCP} \subset \text{SDP} \subset \text{Convex Programs.}$$

Equivalent transformations: why they matter

Many optimization problems do not initially look like standard convex programs.

Main idea

Use variable substitutions, epigraph reformulations, slack variables, and partial minimization to convert them into equivalent convex problems.

1. epigraph form and piecewise-linear minimization
2. linear-fractional programming
3. eliminating / introducing equality constraints
4. slack variables
5. partial minimization and Schur complements

Epigraph form and piecewise-linear minimization

Epigraph form

Any convex problem

$$\min f_0(x)$$

can be rewritten as

$$\min_t \; t \quad \text{s.t.} \quad f_0(x) \le t, \; \text{(original constraints)}.$$

This replaces a nonlinear objective by a linear objective plus one extra convex constraint.

Piecewise-linear objective

For

$$f(x) = \max_{i=1,\dots,m} \{a_i^\top x + b_i\},$$

we solve the LP

$$\min_t \; t \quad \text{s.t.} \quad a_i^\top x + b_i \le t \quad \forall i.$$

- This models worst-case cost over multiple demand or scenario realizations.
- A max of affine functions becomes linear after introducing t.

Why the epigraph reformulation is equivalent I

Consider

$$\min_x f_0(x).$$

Introduce a new variable t and solve

$$\min_{x,t} t \qquad \text{s.t.} \quad f_0(x) \le t.$$

For any fixed x, the smallest feasible value of t is exactly

$$t = f_0(x).$$

So minimizing over (x, t) chooses the same x as minimizing $f_0(x)$ directly.

Geometric view

The feasible set in the new variables is the epigraph

$$\mathrm{epi}(f_0) = \{(x, t) : f_0(x) \le t\}.$$

We are simply minimizing the vertical coordinate t over that set.

Why the epigraph reformulation is equivalent II

A particularly important special case is a maximum of affine functions:

$$f(x) = \max_i \{a_i^\top x + b_i\}.$$

The epigraph constraint $f(x) \leq t$ means

$$\max_i \{a_i^\top x + b_i\} \leq t.$$

That is equivalent to requiring every affine piece to lie below t:

$$a_i^\top x + b_i \leq t \qquad \forall i.$$

So the nonlinear-looking objective becomes a linear objective with linear constraints:

$$\min_t t \qquad \text{s.t.} \quad a_i^\top x + b_i \leq t \ \forall i.$$

This is why worst-case affine objectives naturally turn into LPs.

Chebyshev center and linear-fractional programming

Largest inscribed circle

Given a polytope

$$P = \{x : a_i^\top x + b_i \geq 0\},$$

the Chebyshev center solves

$$\max_{x_c, r} \; r$$

$$\text{s.t. } a_i^\top x_c + b_i \geq r \|a_i\| \quad \forall i.$$

This is an **LP**.

Linear-fractional programming

$$\min \; \frac{c^\top x + d}{e^\top x + f} \quad \text{s.t.} \quad Gx \leq h, \; Ax = b,$$

with $e^\top x + f > 0$.
Use the Charnes-Cooper change of variables

$$y = \frac{x}{e^\top x + f}, \qquad z = \frac{1}{e^\top x + f},$$

to obtain an equivalent **LP**.

Derivation: why the Chebyshev center problem is an LP

Suppose the polytope is described by halfspaces

$$a_i^\top x + b_i \geq 0, \qquad i = 1, \ldots, m.$$

Take a candidate center x_c. The perpendicular distance from x_c to face i is

$$\frac{a_i^\top x_c + b_i}{\|a_i\|_2}.$$

To fit a ball of radius r inside the polytope, that distance must be at least r for every face:

$$\frac{a_i^\top x_c + b_i}{\|a_i\|_2} \geq r.$$

Multiply through by the constant $\|a_i\|_2$:

$$a_i^\top x_c + b_i \geq r \|a_i\|_2 .$$

These are linear inequalities in (x_c, r), so maximizing r is a linear program.

Derivation I: start the Charnes-Cooper substitution

Start with

$$\min_x \frac{c^\top x + d}{e^\top x + f} \qquad \text{s.t.} \quad Gx \le h,\ Ax = b,\ e^\top x + f > 0.$$

Define

$$z = \frac{1}{e^\top x + f}, \qquad y = zx.$$

Then $z > 0$ and

$$x = \frac{y}{z}.$$

Use these substitutions in the constraints:

$$G(y/z) \le h \iff Gy - hz \le 0,$$

$$A(y/z) = b \iff Ay - bz = 0.$$

Derivation II: normalize the denominator and simplify the objective

The denominator becomes a normalization constraint:

$$e^\top y + fz = z(e^\top x + f) = 1.$$

The objective simplifies as

$$\frac{c^\top (y/z) + d}{1/z} = c^\top y + dz.$$

Therefore the equivalent linear program is

$$\min_{y,z} \; c^\top y + dz$$

$$\text{s.t.} \quad Gy - hz \le 0, \; Ay - bz = 0, \; e^\top y + fz = 1, \; z \ge 0.$$

Once (y, z) is found, recover the original variable by $x = y/z$.

Eliminating and introducing equality constraints

Eliminate equalities

If $Ax = b$ and x_0 satisfies $Ax_0 = b$, then every feasible point can be written as

$$x = Fz + x_0,$$

where columns of F span $\mathcal{N}(A)$.
So

$$\min f_0(x) \text{ s.t. } f_i(x) \leq 0,\ Ax = b$$

is equivalent to

$$\min_z f_0(Fz + x_0) \text{ s.t. } f_i(Fz + x_0) \leq 0.$$

Introduce equalities

If expressions $A_i x + b_i$ appear repeatedly, define new variables

$$y_i = A_i x + b_i.$$

Then

$$\min f_0(A_0 x + b_0)$$

can be rewritten as

$$\min f_0(y_0) \quad \text{s.t.} \quad y_i = A_i x + b_i.$$

This simplifies objectives and constraints at the cost of extra affine equations.

Slack variables and partial minimization

Slack variables

An inequality

$$Ax \leq b$$

can be replaced by

$$Ax + s = b, \qquad s \geq 0.$$

This is the standard route from inequality form to equational LP form.

Example:

$$\min 2x_1 + 3x_2$$

$$\text{s.t. } x_1 + x_2 + s_1 = 4, \;\; x_1 + s_2 = 3,$$

$$x_1, x_2, s_1, s_2 \geq 0.$$

Partial minimization

If

$$\widetilde{f}_0(x_1) = \inf_{x_2 : g_j(x_2) \leq 0} f_0(x_1, x_2),$$

then $\widetilde{f}_0$ is convex whenever f_0 is convex and the feasible set for x_2 is convex.

This lets us analytically eliminate variables that only appear in one part of the problem.

Schur complement as an elimination example

Consider the block quadratic form

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}^\top \begin{pmatrix} P_{11} & P_{12} \\ P_{12}^\top & P_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

Minimizing over x_2 gives

$$x_2^* = -P_{22}^{-1} P_{12}^\top x_1,$$

and the reduced objective becomes

$$x_1^\top \left(P_{11} - P_{12} P_{22}^{-1} P_{12}^\top \right) x_1.$$

Schur complement

$$P_{11} - P_{12} P_{22}^{-1} P_{12}^\top$$

is the **Schur complement** of P_{22} in P.

Key fact

If $P \succeq 0$ and P_{22} is invertible, then the Schur complement is also positive semidefinite.

Derivation I: eliminate x_2 by taking a gradient

Expand the block quadratic form:

$$q(x_1, x_2) = x_1^\top P_{11} x_1 + 2x_1^\top P_{12} x_2 + x_2^\top P_{22} x_2.$$

Assume P_{22} is invertible. For fixed x_1, minimize with respect to x_2 by setting

$$\nabla_{x_2} q = 0.$$

Compute the gradient:

$$\nabla_{x_2} q = 2P_{12}^\top x_1 + 2P_{22} x_2.$$

So the minimizing choice is

$$x_2^* = -P_{22}^{-1} P_{12}^\top x_1.$$

Derivation II: substitute back and identify the Schur complement

Substitute

$$x_2^* = -P_{22}^{-1}P_{12}^\top x_1$$

back into the quadratic. After simplification,

$$q(x_1, x_2^*) = x_1^\top \left(P_{11} - P_{12}P_{22}^{-1}P_{12}^\top\right)x_1.$$

The matrix

$$P_{11} - P_{12}P_{22}^{-1}P_{12}^\top$$

is the Schur complement of P_{22} in P.

Why it is PSD when $P \succeq 0$

If $P \succeq 0$, then $q(x_1, x_2) \geq 0$ for all (x_1, x_2). In particular,

$$x_1^\top \left(P_{11} - P_{12}P_{22}^{-1}P_{12}^\top\right)x_1 = \min_{x_2} q(x_1, x_2) \geq 0.$$

So the Schur complement is positive semidefinite.

Solving convex programs with CVXPY

CVXPY in one sentence

CVXPY is a Python modeling language for convex optimization built around **disciplined convex programming (DCP)**.

- DCP encodes composition rules that certify convexity automatically.
- If CVXPY verifies the problem, it can transform it into a solver-ready form.
- Typical back-end solvers include ECOS/SCS for cone programs and OSQP for QPs.

Workflow

1. define variables;
2. define objective;
3. define constraints;
4. build a `Problem` and call `solve()`.

Basic CVXPY pattern

```python
import cvxpy as cp
import numpy as np

x = cp.Variable(n)
objective = cp.Minimize(f(x))
constraints = [A @ x <= b,
               C @ x == d,
               x >= 0]

prob = cp.Problem(objective, constraints)
prob.solve()

print("Optimal value:", prob.value)
print("Optimal x:", x.value)
```

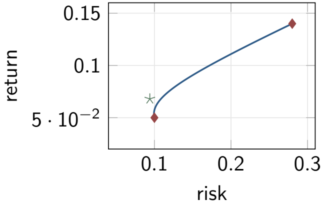
The mathematics is visible directly in code: variables, objective, constraints, solve.

CVXPY example: portfolio optimization

```python
mu = np.array([0.12, 0.10, 0.07, 0.03])
Sigma = ...   # 4x4 covariance matrix

w = cp.Variable(4)
ret = mu @ w
risk = cp.quad_form(w, Sigma)

for r_min in np.linspace(0.03, 0.12, 50):
    prob = cp.Problem(cp.Minimize(risk),
                      [cp.sum(w) == 1,
                       w >= 0,
                       ret >= r_min])
    prob.solve()
```

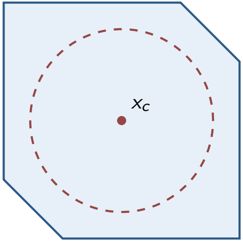


Sweeping the target return $r_{\min}$ traces the **efficient frontier**.

CVXPY example: Chebyshev center of a polytope

```
A = np.array([[ 1, 0], [-1, 0], [0,  1],
              [0, -1], [1,  1], [-1, -1]])
b = np.array([2, 2, 2, 2, 3, 3])
norms = np.linalg.norm(A, axis=1)

x = cp.Variable(2)
r = cp.Variable()
constraints = [A @ x + r * norms <= b,
               r >= 0]
prob = cp.Problem(cp.Maximize(r), constraints
    )
prob.solve()
```



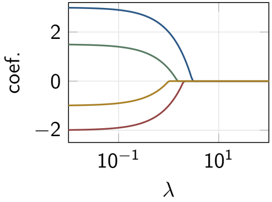
largest inscribed circle

This implements the epigraph/LP reformulation of the largest inscribed circle problem.

CVXPY example: Lasso regularization path

```
beta = cp.Variable(d)
lambd = cp.Parameter(nonneg=True)
prob = cp.Problem(
    cp.Minimize(cp.sum_squares(X @ beta - y)
                + lambd * cp.norm1(beta)))

for lam_val in np.logspace(-2, 2, 100):
    lambd.value = lam_val
    prob.solve()
    # record beta.value
```



As λ increases, coefficients shrink toward zero; this is the sparsity-inducing effect of ℓ_1 regularization.

What the notebook applications illustrate

Applications from the notes

- portfolio optimization
- optimal control
- logistic regression with ℓ_1 penalty
- robust optimization
- Chebyshev center

Pedagogical point

The same mathematical framework appears across finance, control, statistics, and machine learning.

Modeling lesson

Recognizing the problem class (LP/QP/QCQP/SOCP/SDP) often tells you both *how to formulate* the model and *which solver technology* can handle it efficiently.

Summary

- A convex optimization problem minimizes a convex objective over a convex feasible set.
- Standard form uses convex inequalities and **affine** equalities.
- Core structural properties:
  - ε-suboptimal sets are convex,
  - strict convexity implies uniqueness,
  - every local minimum is global.
- For differentiable problems, first-order conditions provide exact global optimality certificates.
- Most important subclasses form the hierarchy

$$\mathrm{LP} \subseteq \mathrm{QP} \subseteq \mathrm{QCQP} \subseteq \mathrm{SOCP} \subseteq \mathrm{SDP}.$$

- Equivalent transformations let us convert many problems into standard convex forms.
- CVXPY makes practical convex modeling concise and readable.